Jiachen Qian

M.S. Student in Computer Science, City University of Hong Kong

Ph.D. Applicant in Trustworthy Multimodal AI and Agent Security

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Research Profile

Computer science M.S. student at the City University of Hong Kong, working on trustworthy agentic systems and generative AI security. My first-author work studies how multimodal perception, long-term memory, and external knowledge retrieval act as stealthy attack surfaces in e-commerce agents, recommender systems, and RAG pipelines. Current work includes four first-author papers accepted/submitted, including two ACL 2026 papers, one KDD 2026 paper, and one ECCV 2026 submission; one ACL 2026 paper and the KDD 2026 paper are sole-author works. I also co-authored an arXiv preprint on LLM order sensitivity.

Research Interests

Trustworthy agentic systems; generative AI security; e-commerce, recommender, and RAG systems; adversarial machine learning; memory poisoning and defense; LLM behavioral analysis; reliable decision-making.

Publications and Manuscripts

First-Author Papers

- **Qian, Jiachen** and Zhaolu Kang. "Penny Wise, Pixel Foolish: Bypassing Price Constraints in Multimodal Agents via Visual Adversarial Perturbations." *Findings of ACL 2026*, accepted/to appear; arXiv:2604.16515.
  First author; formulated the multimodal shopping-agent threat model, developed PriceBlind, and led evaluation and defense analysis.
- **Qian, Jiachen**. "Visual Inception: Compromising Long-term Planning in Agentic Recommenders via Multimodal Memory Poisoning." *ACL 2026 Main Conference*, accepted/to appear; arXiv:2604.16966.
  Sole author; characterized multimodal long-term memory as a stealthy attack surface and proposed a prompt-free visual memory-poisoning attack with the CognitiveGuard defense.
- **Qian, Jiachen**. "SilentRetrieval: Hijacking Retrieval-Augmented Generation via Semantically-Preserving Adversarial Data Poisoning." *KDD 2026*, accepted/to appear.
  Sole author; proposed SilentRetrieval, a stealthy corpus-poisoning framework for RAG that jointly manipulates retrieval and downstream generation.
- **Qian, Jiachen**, et al. "Security of Large Models for Tampered-Image Detection in Social Media." *ECCV 2026*, under review.
  First author; led attack/defense code, soft-attention analysis, and iterative prompt optimization for the attack pipeline.

Co-Authored Work

- He, Yingjie, Zhaolu Kang, Kehan Jiang, **Qian, Jiachen**, et al. "How Order-Sensitive Are LLMs? OrderProbe for Deterministic Structural Reconstruction." *arXiv preprint* arXiv:2601.08626, 2026.
  Co-author; LLM order sensitivity and deterministic structural reconstruction.

Research Experience

Security of Multimodal Shopping Agents — Hong Kong SAR

Graduate Research Project, City University of Hong Kong — *Sep 2025 – Present*

- Studied reliable decision-making in multimodal shopping agents on e-commerce platforms, focusing on how visual signals can override explicit textual price evidence under goal constraints.
- Identified the Visual Dominance Hallucination failure mode and developed PriceBlind, a stealthy attack framework that preserves pixel-level fidelity while systematically bypassing "buy-the-cheapest"-style constraints.
- Evaluated attacks on E-ShopBench, achieving about 80% white-box attack success and 35–41% transfer success across GPT-4o, Gemini-1.5-Pro, and Claude-3.5-Sonnet.
- Assessed practical defenses including Verify-then-Act and robust visual encoders, connecting attack behavior to agent decision reliability.

Security of Memory-Augmented Recommendation Agents — Hong Kong SAR

Graduate Research Project, City University of Hong Kong — *Sep 2025 – Present*

- Investigated system-level vulnerabilities in long-horizon recommender agents, characterizing multimodal long-term memory as a stealthy attack surface that can remain dormant and later influence downstream planning.
- Designed a prompt-free visual memory-poisoning attack that implants dormant cues into user-uploaded images and later hijacks recommendation planning and goal selection.
- Proposed CognitiveGuard, a dual-process defense combining diffusion-based sanitization with counterfactual consistency checks to mitigate persistent memory contamination.
- Reduced Goal-Hit Rate from about 85% to about 10% in evaluation, with configurable trade-offs between latency and defense strength.

Research Experience

RAG Security and Semantically Preserving Corpus Poisoning Hong Kong SAR

Graduate Research Project, City University of Hong Kong *2026*

- Studied the security of retrieval-augmented generation, characterizing external knowledge corpora as a stealthy attack surface capable of influencing both retrieval and downstream generation.
- Proposed SilentRetrieval, a semantically preserving corpus-poisoning framework that combines Coordinated Beam Search (CBS) and Context-Adaptive Trigger Generation (CATG) to jointly control retrieval hits and generated answers without degrading text naturalness.
- Achieved 84.6% / 81.3% HR@10 and 57.5% / 54.8% ASR-LLM on NQ and MS MARCO, showing stable attacks on both retriever ranking and answer generation.
- Evaluated reranking, hybrid retrieval, and passage isolation as defenses, analyzing their trade-offs between robustness and practical utility.

Security of Large Models for Tampered-Image Detection in Social Media Shenzhen, China

Collaborative Research Project, Shenzhen MSU-BIT University *2026*

- Studied the security of large-model-based tampered-image detection in realistic social-media settings, focusing on adversarial vulnerabilities and reliability limits of recognition-oriented multimodal models.
- Led attack/defense code design and implementation, using soft-attention behaviors in large models and iterative prompt optimization to build the attack pipeline, and carried out experimental evaluation and reliability analysis for a collaborative ECCV submission under review.

Education

City University of Hong Kong Hong Kong SAR

Master of Science in Computer Science *Sep 2025 – Jun 2027 (expected)*

- Research focus: trustworthy agentic systems and generative AI security, with emphasis on stealthy attack surfaces in multimodal perception, long-term memory, and external knowledge retrieval.

Nanjing University of Information Science and Technology Nanjing, China

B.Eng. in Software Engineering *Sep 2021 – Jun 2025*

Patent

- "A Distance Measurement Device Based on Computer Vision." Invention patent, No. 202310436071.5.
  Computer-vision-based distance measurement device.

Academic Service

- Reviewer for *IEEE Transactions on Information Forensics and Security (TIFS)*.

Honors and Awards

- Outstanding Graduate at both school and university levels, Nanjing University of Information Science and Technology, 2025.
- First-Class Scholarship, 2021 – 2022.
- Third-Class Scholarship, 2022 – 2023; Third-Class Scholarship, 2023 – 2024.

Technical Skills

- Agent Security: attacks and defenses for e-commerce agents, recommender systems, and RAG pipelines.
- LLM/VLM Evaluation: multimodal reliability analysis, soft-attention analysis, iterative prompt optimization, benchmark construction.
- Research Engineering: PyTorch, CUDA, Python, C++, SQL, Git, LaTeX, automated and reproducible experiment pipelines.